From AI4SE Pilot to Conditional Scale: A Transformation-Path Experience in Mid-to-Large R&D Organizations

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***Abstract*—AI for Software Engineering (AI4SE) uses AI in software-engineering work. Here, "mid-to-large" means more than 200 R&D staff; two client cases, Org-A and Org-B, each have more than 1,000 R&D staff, multi-site delivery, brownfield systems, and established processes. In this setting, heterogeneous AI practice creates verification and transfer risk, while sponsors may mistake a pilot for scale readiness. Existing staging vocabulary does not specify how to separate pilot-method closeout from wave-specific Scale admission. This study addresses that problem by reporting an evidence-based transformation framework and method organized as an Explore–Consolidation–conditional-Scale path, with evidence reviews. Results show that, at Org-A, a six-week Explore produced citable method, opportunity assets, and a readiness-labeled executable ledger; management passed Explore closeout but withheld Scale admission. In an observed five-week Consolidation slice, the team recorded 15 requirements, 18 changes, and 8 finished changes. All 18 changes were processed through the Spec-Driven Development (SDD) workflow, with a corresponding specification produced for each change; Verify, Align, and role gaps kept Scale conditional. After judging alternative AI4SE approaches infeasible for its setting, Org-B selected the path instantiated at Org-A and began adapting it to incumbent IPD and SAFe-class constraints. Its record is a briefing-level adaptation design, not evidence of formal closeout, later-stage delivery, or an ART/Portfolio rewrite. Findings support a bounded feasibility argument with unequal evidence strength: Org-A supplies observed method-chain and Consolidation evidence, while Org-B supplies an adaptation design. The contribution separates pilot closeout from conditional Scale admission: completion can trigger evidence review and support Consolidation, but not Scale without transfer evidence. We do not claim organization-wide Scale, completed Scale admission, or platform-measured return.**

***Index Terms*—AI for software engineering, AI4SE transformation path, industrial experience report, mid-to-large R&D organizations**

I. Introduction

In this paper, "mid-to-large" means an organization with more than 200 R&D staff. The two client organizations examined here, Org-A and Org-B, each have more than 1,000 R&D staff, operate across multiple delivery sites, maintain substantial brownfield systems, and rely on established, relatively conventional engineering processes. In these mid-to-large engineering organizations, AI assistants are adopted unevenly across tools, prompts, specification habits, and human–agent handoffs. Heterogeneity raises organizational entropy: process risk grows, quality is hard to underwrite when verification and intent ownership fragment, and key-person departure removes tacit "AI-fluent" know-how that never entered a citable method. Enterprise software engineering must therefore pursue *trustworthiness* and *sustainability*, not throughput slogans alone [1]. The motive continues earlier industrial process change: large manufacturing organizations already used process-and-quality transformations such as Integrated Product Development (IPD) to keep complex integration credible; under AI, AI4SE is the analogous upgrade for human–agent delivery—speed subordinated to governable evidence. The paper targets complex-delivery organizations: small innovative firms may prioritize speed over heavy gates, whereas organization-wide "AI acceleration" requires management capacity and maturity that many incumbents lack.

Sponsors still err by equating assistant access, or a facilitated pilot, with scale-ready capability. Neither shows that a new team can run a complete cycle under agreed guardrails. AI assistance compresses *implementation* cost while amplifying errors from vague intent, weak specifications, and unverified agent output (*definition scarcity* and verification capacity). Treating a well-received pilot as organization-wide diffusion authorization is the failure mode this paper records.

Five recurring AI4SE failure modes (DevOps patterns reframed for AI) also shape the design: *value-curve misread* (underestimating the transformation J-curve: early gains can be followed by a temporary efficiency dip as teams absorb new practices, add testing, and pay down technical debt [2]; in AI4SE, learning to validate and repair model output can deepen that dip); *no pilot capacity* under delivery pressure (fragmentary tool trials); *tool-procurement equivalence* without redesigning review/security/compliance for generated artifacts; *external-consultant dependence* without developing internal coaches who understand both software engineering and model limitations; and *metric mismatch* (LOC/hours/speed rewarding unreviewed AI paste-ins). The path counters them with evidence-based admission, method-before-tool, teachable ownership, and measurement on a companion assessment track.

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The unresolved bottleneck is therefore not whether AI can assist one team, but how an organization can decide when a locally feasible method has become transferable to another team.

We propose and report an evidence-based AI4SE transformation framework and method. Its *path* is Explore–Consolidation–Scale; its *admission contract* specifies stage-boundary evidence; and its *operating stack* names the delivery vocabulary carried by the path. Relative to classical Stage-Gate business-case gates [3], admission inspects a citable method, a stub-versus-ready ledger, and transfer preconditions. Each stage has a purpose, work products, and an observable exit that tests entry to the next: Explore asks whether a method runs end-to-end, Consolidation whether the pilot team can repeat and teach it, and conditional Scale whether a new team can run a cycle independently. Org-A supplies the primary longitudinal evidence. After Org-B judged alternative AI4SE transformation approaches infeasible for its setting, it selected the transformation path instantiated at Org-A and began adapting it to its IPD and SAFe-class constraints; the available record is a briefing-level late-Explore adaptation design. The transferable artifact is therefore the framework's path-plus-admission contract, not a tool catalog or completed rollout; operating-stack terms name the practices carried by that contract.

We organize the experience around three questions. The questions are stated at the level of the framework and its evidence; Org-A and Org-B are the evidence sites that answer them, not part of the question wording:

- **Q1.** What evidence-based framework and method can structure an industrial AI4SE transformation from pilot exploration toward conditional scale, so that calendar completion is not treated as scale readiness? (Answered in Section III.)
- **Q2.** What evidence can a completed Explore produce that supports Explore closeout and entry into Consolidation without supporting Scale admission? (Answered in Section IV, primarily at Org-A.)
- **Q3.** What team-level adaptation design was documented for the SDD-centered framework in late Explore under incumbent IPD and SAFe-class constraints, and what evidence boundaries remained? (Answered in Sections V and VI, primarily at Org-B.)

Methodological stance. This manuscript is an industrial experience report [4]. Facilitators co-designed the path, but the defended claims are Org-A's management decisions (Explore closeout met / Scale admission held) and the readiness labels recorded at the executive readout, not a consultant self-score. Org-B is a secondary late-Explore briefing case with no intranet mirror. We do not claim theoretical saturation, a completed DSR evaluate cycle [5], researcher-owned action-research agency, or completed Scale at either organization.

Contributions. (1) an evidence-based framework and method for mid-to-large organizations, with an Explore–Consolidation–Scale path, stage work products, a citable/runnable/transferable admission contract, and a compact operating stack; (2) bounded Org-A feasibility evidence—Explore closeout, an observed client-operated Consolidation midpoint, and withheld Scale admission; and (3) a briefing-level Org-B account of team-level adaptation under harder IPD plus SAFe-class constraints, without evidence of formal closeout, Consolidation, or Scale admission. Both organizations plan subsequent scale work, but that plan is not organization-wide Scale. Explore effectiveness and maturity signals are bounded stage inputs; their quantitative stratification belongs to a companion manuscript and is not reproduced here.

Section II supplies background vocabulary. Section III defines the stages and closeout/admission criteria. Section IV reports Org-A. Section V reports Org-B. Sections VI–IX discuss takeaways, threats, related work, and the claim ceiling.

II. Conceptual Background

This section gives compact vocabulary for why adoption is staged and what the path carries (an operating-stack reading). The vocabulary is background, not a second framework to evaluate and not a slide-deck restatement.

A. Domain Fit: Why Exploration Is Not Instant Scale

Snowden and Boone's Cynefin framing distinguishes decision domains in which best practice, expert analysis, or probe-and-sense behavior is appropriate [6]. Early industrial AI4SE work is typically *complex*: workable method is emergent, so Explore is a probe that must produce inspectable method assets rather than a purchased best-practice pack. Consolidation moves durable findings toward the *complicated* domain: playbooks, harness hardening, and seed coaches support sense–analyze–respond transfer. Scale entry is admitted only when enough of the method is stable to diffuse without re-running full discovery—closer to clear or well-analyzed complicated work, never by calendar alone. The gates therefore match the decision to the nature of the work. A method still being explored in the Complex domain should not be treated as ready for organization-wide rollout. The failure mode recorded here is pilot success without Scale admission.

B. Industry Fault Lines as Staging Rationale

Senior-practitioner syntheses from Thoughtworks' Utah and Europe Future of Software Engineering retreats highlight fault lines that motivate staging [7], [8]: engineering rigor migrating into specifications, tests, and constraints; a supervisory *middle loop* between coding and delivery; verification (not generation) as scarce capacity; harness ownership around the agent; and a two-clock mismatch in which code-production time collapses while decision time does not. We use that map only as industrial *rationale*: Explore freezes probe evidence; Consolidation hardens verification, harness ownership, and seed coaches; Scale waits on decision/verification capacity for a receiving team. Retreat anecdotes are not Org-A/Org-B facts (Section VIII).

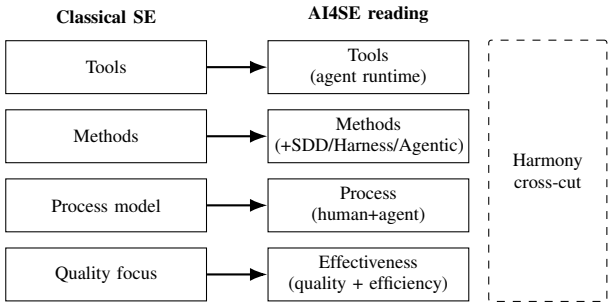


Fig. 1. Inherited layered software-engineering structure with upgraded AI4SE semantics (conceptual). Harness/SDD/Agentic sit primarily in Methods; Harmony cross-cuts rather than stacking a fifth layer.

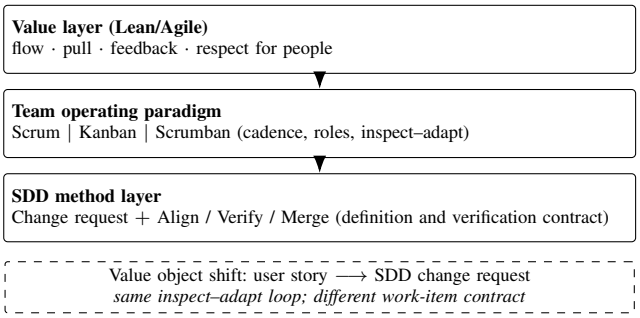


Fig. 2. Three-layer AI4SE operating stack. Classical Lean/Agile team modes remain; the primary value-flow object moves from stories toward SDD change requests.

C. Layer Inheritance: Pressman Structure, Upgraded Semantics

Classical software-engineering textbooks present a layered view in which a quality focus grounds process, methods, and tools, and in which more visible layers depend on less visible ones [9]. Figure 1 states our industrial reading: the *structure* is inherited; the *semantics* must be upgraded for human–agent delivery. Quality focus becomes *Effectiveness* (quality—including *trust*—plus waste-aware efficiency that serves *value*, not speed alone). Process, methods, and tools retain their roles, but executors are humans *and* agents. Harness Engineering, Spec-Driven Development (SDD), and Agentic Engineering are placed in the *methods* family (discipline and assets), not as a replacement for the process layer or as a tools-only purchase. *Harmony* is a cross-cutting human–agent contract, not a fifth stacked layer.

D. Operating Stack: Value Layer, Team Paradigm, and SDD

Pressman's layers answer *what* to upgrade; delivery teams still need an *operating stack* that says how Lean/Agile work continues under AI4SE. Figure 2 states three nested layers we used with sponsors and coaches. The top layer keeps Lean/Agile value principles (flow, pull, feedback, respect for people). The middle layer keeps classical team operating modes—Scrum, Kanban, or Scrumban—as the cadence and role container (Org-A primarily Scrum; Org-B combines IPD and SAFe-class scaled-agile constraints with multiple team modes). The bottom layer is Spec-Driven Development (SDD): Align, Verify, and Merge reviews that protect each change under the chosen paradigm p.

The industrial claim is deliberately narrow: those team modes remain effective; what shifts is the primary *unit of value flow*. Where boards and backlogs once centered on user stories, SDD centers on a citable *change request* (and its Align→Verify→Merge contract) as the object that carries definition and verification evidence. Stories and other backlog items may still exist; they no longer substitute for an inspectable change contract when agents draft and apply increments. As implementation cost falls, definition and verification become the scarce steps within the same inspect–adapt container (a gradual shift toward change-request flow, not a discard of Scrum cadence). We reserve the word *gate* for those process-level Align/Verify/Merge reviews in the method pack, and we use *closeout* / *admission* for organizational stage decisions in Section III—so the two loops do not reuse the same numbered gate labels.

E. Effectiveness, Harmony, and Four In-Path Directions

We use two cross-cutting concerns—*Effectiveness* and *Harmony*—and four stage-relative directions (SDD, Agentic Engineering, Harness Engineering, Operating Model) as path vocabulary, not as a maturity model or a closeout/admission substitute. Effectiveness requires inspectable evidence that selected work ran under local constraints (citable method, validation-vehicle run-through, or new-team cycle) rather than attendance; quality includes *trust*, and efficiency means valuable outcomes without waste. Harmony requires human–agent accountability under four named functions—*Intent Owner*, *Spec Steward*, *AI Orchestrator*, and *Verification Lead*—which may be combined but not vacant, with *contract separation* forbidding the apply party as sole verifier. As enablement heuristics (not measured Org-A/Org-B protocols), formation change stays minimal (Scrum-class cadence; Spec Steward + Verification Lead cells; shared Intent Owner / Orchestrator only when apply is not self-certified), and risk routing uses HITL / HOTL / HOOL by risk band. Along the path: **SDD** yields a citable then teachable specification contract; **Agentic** work must become a guarded main path; **Harness** moves from disclosed scaffold to hardened starter kit (stub $\neq$ ready; Section IV); **Operating Model** names owners and seed coaches for a receiving team. SDD/agentic writing [10] is peer context only.

F. Certainty Boundary: User Harness, Model, and Method Assets

Industrial harness writing treats the coding-agent harness as the non-model layer (guides, sensors, and user-owned outer controls) in a bounded context [11]. Figure 3 is our AI4SE reading: organizational certainty does not live in the model alone. A governing delivery paradigm p (for example SDD versus loosely guided "vibe coding") selects outer user-harness and coding-agent contracts around a comparatively stable model; Explore must freeze inspectable structure and Consolidation must harden ownership. Tool design is *process-shaped*: AI assets (commands, agents, skills, rules, hooks)

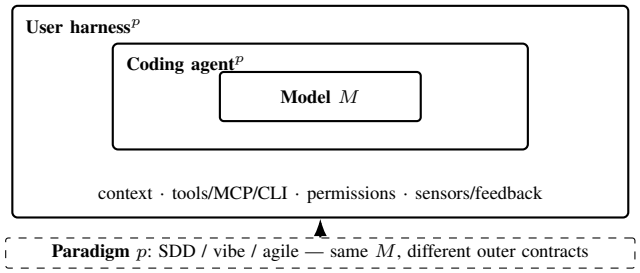


Fig. 3. Certainty boundary for AI4SE enablement (conceptual adaptation of harness-as-non-model-layer thinking [11]). Nested layers around model M; paradigm p selects outer contracts. Agentic engineering composes them—it is not a competing fourth paradigm.

package against Align/Verify/Merge and stage flows under a User Harness Framework (core workflow $\leftrightarrow$ user-asset pack), not a post-hoc tool catalog; *agentic engineering* composes them into a governed main path, with HITL/HOTL/HOOL duty inside asset design. Method assets feed the harness; they do not replace the model. The harness remains an open SDD-stack class (OpenSpec / Superpowers / Spec-Kit style examples, and peers)—a constraint Org-B makes explicit in Section V.

Skipping Consolidation typically leaves SDD prose and a stubbed harness with a new team—the failure mode Section III names as equating a pilot with scale.

III. Adoption Rhythm: Explore to Conditional Scale

A. Why Closeout and Admission, Not a Calendar, Advance the Path

We define an AI4SE transformation path as a sequence of capability-building stages separated by explicit management decisions. A stage has a purpose, work products, and an observable exit condition. Its exit condition is also an admission test for the next stage: elapsed time can trigger a review, but cannot substitute for evidence. This distinction is important because a pilot can finish on schedule while leaving its method tacit, its automation fragile, or its ownership unresolved. Collapsing code-production time does not by itself collapse decision and verification time—another reason reviews must examine evidence rather than calendars.

The transformation path nests two control loops. *Stage closeout and admission decisions* decide whether the organization may enter Consolidation or admit a Scale wave. Inside delivery work, shorter *Align/Verify/Merge* reviews—align a delta specification, verify a candidate increment, then merge and archive—protect each change under the chosen paradigm p. Stage criteria ask whether method, assets, and ownership are transferable; Align/Verify/Merge reviews ask whether a particular change is definition-complete and evidence-backed. Confusing the two is a common failure: a team that passes local change reviews can still fail Scale admission if receiving teams cannot inherit the method. (Process vocabulary only: Align/Verify/Merge sit mainly in local coding/harness loops; stage closeout/admission remain outer governance.)

The path instantiated at Org-A comprises *Explore*, *Consolidation*, and *Scale* (Fig. 4), with the stage questions stated in Section I. Outputs flow forward as dependencies: the first-generation playbook and pilot assets feed hardening; the hardened method, executable starter kit, role owners, and seed coaches then support controlled diffusion.

B. Stages, Deliverables, and Exit Criteria

Stage 1—Explore. The purpose of Explore is feasibility, not broad deployment. In the Org-A instantiation, the indicative duration was approximately six weeks, spanning diagnosis and scenario selection, four workshop weeks over fourteen opportunity themes, co-creation of an initial playbook and executable-asset scaffold, and use of a pilot change request as a validation vehicle. The exit criterion was an end-to-end run-through of the method chain. It was deliberately not completion of every feature in the validation vehicle.

Stage 2—Consolidation. Consolidation is the bridge between one successful run and a transferable organizational capability. Its indicative Org-A window was one to two months. The work comprises revising clauses that read correctly but do not run in practice, hardening the principal executable path from stubs to usable assets, assigning role ownership, and rehearsing coach-the-coach transfer. Its intended products are a standardized playbook v2, a hardened toolset, and seed coaches. Hardening follows two practical rules drawn from the enablement practice: *do not promote immature automation*, and *keep apply separate from verify*. As a Consolidation heuristic we refuse to skip rule $\rightarrow$skill $\rightarrow$agent $\rightarrow$command $\rightarrow$hook promotion; we do not treat reuse/hit/drift/retire rates as ROI here. The exit condition is "team can teach": pilot-team members can explain and transfer the method, rather than only reproduce a consultant-led demonstration.

Stage 3—Scale. Scale is quality-consistent, wave-based diffusion rather than simultaneous organization-wide rollout. Admission requires the Consolidation exit, named role owners, a starter kit for sibling teams, and reserved sponsor cadence and resources. Seed coaches then enable successive teams and review practice consistency. The terminal evidence is a fresh team's independent completion of a cycle under the guardrails. Thus, a calendar target can prompt a Scale-admission review, but unmet criteria block admission.

C. Closeout and Admission Semantics

Explore closeout is a *method–asset closeout*. Management asks whether a playbook can be shown and cited with its gaps visible; whether opportunity-point specifications and best practices can be inventoried; whether executable assets are distinguished as ready or stub; whether effectiveness and maturity signals carry their measurement boundaries; and whether the Consolidation-to-Scale path is concrete enough to discuss. A positive Explore closeout therefore licenses method hardening, not broad rollout. Table I makes this decision contract explicit.

Scale admission is stricter because it concerns transfer beyond the pilot setting. It tests whether know-how has moved

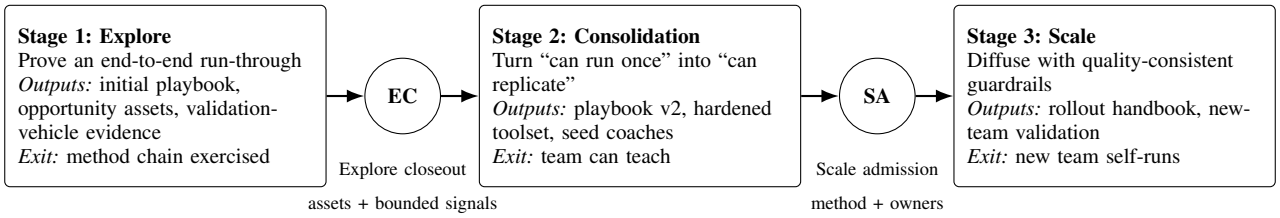


Fig. 4. Explore–Consolidation–Scale adoption rhythm with Explore closeout (EC) and Scale admission (SA) criteria. The figure specifies evidence required to advance; it does not indicate that Org-A completed Scale.

TABLE I
DECISION TESTS AT THE TWO STAGE BOUNDARIES

Boundary	Evidence reviewed	Positive decision	What the decision does not imply
Explore closeout	Citable playbook with gaps stated; inventory of converged and observed opportunity assets; readiness of executable assets stated as ready or stub; bounded effectiveness/maturity signals; discussable next-stage path	The method chain ran end-to-end and the organization may enter Consolidation	All validation-vehicle features are complete; executable assets are ready for broad use; measured organization-wide return has been established
Scale admission	Team-can-teach evidence; hardened main execution path; named role owners; starter kit; sponsor rhythm and resources	A controlled wave may admit a new team with seed-coach support	The Explore pilot alone authorizes diffusion; every team is already independent; rollout outcomes are known in advance

from individuals into teachable guidance, executable assets, role ownership, and an enablement mechanism—and whether verification authority remains human-held under the nested Align/Verify/Merge reviews. The resulting admission decision is reversible and wave-specific: if a new team cannot operate with the starter kit and bounded coaching, the organization returns the deficient asset or role arrangement to Consolidation. This interpretation makes a stage decision a learning control, rather than a ceremonial approval checkpoint.

D. Operating Stage Criteria as an Evidence Review

A usable stage review needs a review packet, a decision owner, and a recorded disposition. The review packet should present the stage goal first and then map each promised output to an inspectable artifact or observation. A readiness label accompanies each item so that "exists," "runs on the primary path," and "transferable to a new team" cannot collapse into one category. Known gaps remain in the packet as decision inputs. Removing them from view would make an approval easier to obtain but less useful for planning the next stage.

The decision owner evaluates sufficiency for the *next* stage, not perfection in the current one. At Explore closeout, an incomplete executable scaffold can be acceptable when it is disclosed and when the Consolidation plan names the hardening work. At Scale admission, the same scaffold is a blocker because a receiving team would depend on it. This relative interpretation prevents two common errors: rejecting useful Explore evidence because every asset is not yet finished, and admitting Scale because unfinished assets were tolerated at Explore closeout.

A stage disposition has three possible forms. *Advance* means the exit condition is met and the next stage may start under its declared scope. *Advance with conditions* records a narrow action whose completion is checked before a dependent activity begins; it must not be used to waive a core exit condition. *Hold* returns named gaps to the current stage. In all three forms, the evidence and rationale remain available for the next review. This creates continuity across decisions and makes negative evidence—for example, a stubbed primary path or missing owner—actionable rather than embarrassing.

The review also separates two kinds of uncertainty. Product uncertainty asks whether AI4SE can improve the selected work under local constraints; Explore reduces it through actual method use. Transfer uncertainty asks whether another team can reproduce that use without recreating the method; only Consolidation and a new-team test can reduce it. Treating these uncertainties separately explains why a positive Explore closeout and a negative Scale admission can be the expected, internally consistent outcome of the same review.

E. Three-Claim Path-Progression Argument

The path permits a useful feasibility judgment before claiming completed diffusion. We structure that judgment as three claims (not process L0/L1 levels, and not the Method–Evidence asset classes).

Claim 1—the path is designed. Stage goals, work products, exits, and admissions are stated before scale readiness is assessed. This makes the claim falsifiable: missing ownership, hardening, or transfer evidence can stop advancement even when a pilot was well received.

Claim 2—Explore is observed end-to-end. For Org-A, the initial playbook, opportunity assets, and validation vehicle exercised the method chain under real delivery constraints. The evidence supports "can run once" and justified meeting Explore closeout; it does not establish repeatability across teams.

Claim 3—Scale remains conditional. Org-A's pilot succeeded at Explore, but Scale admission was not met at close-

out. Consolidation—including asset hardening and team self-solve capability—is the prerequisite for a later Scale decision. The feasibility judgment therefore rests on an explicit path, observed end-to-end use, a readiness-labeled asset ledger, and testable admission conditions. It does not rest on an assumed organization-wide return or on diffusion that has not occurred.

F. Anti-Patterns Exposed by the Criteria

Tool-only transformation treats access to a coding assistant or plugin catalog as the change itself. It omits the specifications, verification, context, roles, and learning routines needed to make tool use repeatable. Explore closeout exposes this weakness by requiring a citable method and bounded evidence; Scale admission exposes it by requiring transfer and ownership.

Pilot equals scale converts a successful demonstration directly into rollout authorization. It conflates end-to-end feasibility with cross-team replicability. Separating Explore closeout from Scale admission preserves the useful result of the pilot while making the missing admission evidence visible.

Skipping Consolidation carries first-generation prose and stubs directly into new teams. The likely burden then moves from explicit hardening to repeated local improvisation. The Stage-2 exit—team can teach—requires the organization to remove that burden before opening a Scale wave.

Agent self-certification lets the same generative path that applied a change also declare the change verified. Nested Align/Verify/Merge reviews and the Harmony verification function exist to forbid that shortcut: drafts and execution may be agentic; sign-off remains human.

IV. Case Org-A

A. Industrial Context and Evidence Basis

Org-A is an anonymized large global container-shipping and logistics enterprise with a substantial, multi-site in-house IT organization supporting commercial and operational systems. The engagement was an approximately six-week AI4SE enablement pilot, not a scale-out program. The delivery setting used a multi-team agile model, shared requirements management, and distinct development, business-analysis, and quality roles. Quality remained a standing constraint: efficiency exploration was not allowed to relax the client's existing quality controls.

We report the case as an industrial experience and preserve a chain from each case statement to a frozen evidence slice, following the traceability principle in case-study reporting [4]. The primary case record comprises the phase timeline, the stage-criteria definition, the Method–Evidence handoff ledger, and post-closeout consolidation signals. Org-A is the primary evidence instantiation of the proposed framework. We treat the playbook and executable assets as designed artifacts evaluated through use in their organizational environment, consistent with design-science evaluation logic [5].

Engagement was *method-led* (not tool-procurement-led): a self-developed AI4SE method guided tool fit, process redesign, and human–agent role boundaries, with selected workflows encoded into the harness/platform as they hardened. Executable encoding was still incomplete at Explore closeout (main chain largely stubbed); Consolidation continued it. Org-A also stood up a tailored AI4SE maturity model and DORA-aligned scaffolds (Cycle Time, Throughput) for later conditional Scale, including differentiated team targets (maturity-score bands; Cycle Time $\sim$1 vs $\sim$2 weeks by readiness) as a Scale-prep pattern. Numeric maturity/DORA results remain the companion measurement study's unit; this path paper uses only qualitative signal boundaries.

B. Explore Activities

The Explore phase followed a six-block arc over approximately six weeks (Fig. 5). Kickoff established the diagnosis, scenario lock, baseline maturity exercise, and opportunity-theme plan. Workshop W1 addressed engineering foundations: harness/context, knowledge, and brownfield specification-driven development. W2 covered metrics and requirements entry and included a midterm management showcase. W3 addressed review, agent validation, and test engineering. W4 ran two tracks in parallel: playbook v0.5 authoring and specification-based validation in the delivery track. An executive readout at the end of Explore closed the phase, aligned the Consolidation path, and explicitly did not claim Scale admission.

Across those four workshop weeks, the team worked through fourteen opportunity themes—twelve numbered plan themes plus two co-created themes—and co-created initial method and opportunity assets. A pilot change request supplied the validation vehicle for using the method in delivery. This choice kept acceptance focused on whether the chain from opportunity through specification and practice could be exercised, rather than on shipping all business scope. The activity record consequently supports a process-feasibility conclusion, not a claim about broad adoption.

Opportunity co-creation followed Design Thinking's Double Diamond (diverge–converge) rather than a consultant-authored theme list [12]: facilitators guided participants to brainstorm and converge *with* AI assistants, and converged outputs seeded Opportunity-layer specifications (and later best practices). We report this as an Explore enablement mechanism, not a controlled Design Thinking or AI-brainstorming evaluation.

C. From Activities to Stage Evidence

The workshop sequence was not treated as attendance evidence. Each block was expected to leave a trace in one or more asset layers: foundation work fed method guidance and executable scaffold; Double Diamond opportunity work created specifications and best practices; delivery-track use tested whether those descriptions could guide action; and the closeout pack recorded signals and their limits. W4's parallel playbook authoring and specification validation formed the key feedback loop—delivery exposed weak wording or stubs; the playbook remained a revisable, citable object—so closeout assessed artifacts and gaps rather than inferring capability from workshop completion. We used three readiness distinctions:

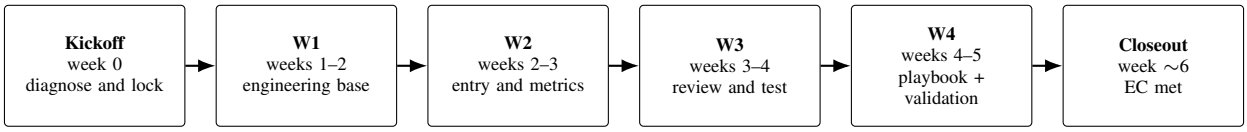


Fig. 5. Org-A Explore timeline (relative week windows across an approximately six-week enablement). The closeout review met Explore closeout and opened Consolidation; it did not meet Scale admission.

citable (stable enough to discuss), *runnable* (principal path works locally), and *transferable* (another team can use it under declared guardrails). Explore required a citable method and local use; Scale admission would require runnable and transferable support—hence a Method-layer playbook could meet Explore closeout while a largely stubbed Executable pack could not meet Scale admission.

D. Method–Evidence Outcomes and Readiness

The closeout inventory was organized into four asset classes (Table II): *Method*, *Opportunity*, *Executable*, and *Evidence*. We avoid L1–L4 labels so they are not confused with process hierarchy (e.g., L0/L1) or Level/Layer abbreviations. The classes distinguish descriptive guidance, reusable opportunity knowledge, executable support, and evidence. The ledger records readiness separately from existence; a catalog entry is not treated as a deployable capability.

Method layer: citable method. The frozen v1.0 handbook captured the end-to-end method—including how intent becomes living specifications, change deltas, and accepted increments—well enough to be shown and referenced at closeout. Its frozen state made the Explore result auditable; it did not mean that Consolidation revisions were complete. Playbook v2 remained a Stage-2 target. A public method card (chapter map, five first principles, Living-Spec chain, 5+1 delivery stages 0–5 from baseline through archive/evolve with nested Align/Verify/Merge reviews, paraphrased clauses, and declared v1 gaps) is released with the data pack;[1] the full handbook is not. As an illustrative Method-layer extract (paraphrased): v1.0 required every accepted increment to cite a living-specification delta id, name a human verifier distinct from the apply agent, and refuse "generated-output equals verified" shortcuts. Org-A's Consolidation agenda explicitly included rewriting Executable-main-chain clauses that "read right but did not run." Non-ROI intensity at closeout was visible without claiming return: four workshop weeks plus kickoff/readout, fourteen themes, 11+13 Opportunity-layer assets, a largely stubbed Executable main chain, and Explore closeout met / Scale admission held.

Opportunity layer: countable opportunity knowledge. The release set contained 11 best practices and 13 specifications across fourteen themes (twelve numbered plan themes plus two co-created: agent architecture; SDD Scrum events); the named index is in the public `theme-index.csv`. Counts are not 1:1 theme–spec–practice: some themes share a specification, theme #1 contributes two, and two release specifications sit beside the main index. Themes #5 and #9 remained observation items (no release best practice), so management could count the handoff without erasing uncertainty.

Executable layer: scaffold, not production readiness. The handoff contained a formal executable-asset pack whose redacted layout was: `commands/` (`/sdd` catalog), `skills/{intent,spec,design,verify,harness}/`, `recipes/` (ordered stage sequences), `adapters/` (coding-agent hosts), and `agents/` (sample wrappers only). The catalog existed, but the main execution chain was largely stubbed: many skill bodies were scaffolds, and stage recipes were not yet a hardened primary path. Org-A therefore recorded the assets as present but did not infer production readiness from the directory tree, and did not treat early agent wrappers as mature automation. Moving the primary chain from stub to ready was explicitly assigned to Consolidation.

The validation vehicle was a pilot change request on a live delivery track: the team exercised intent capture, living-specification update, increment application, and human-owned verification under retained quality controls. Acceptance for Explore stayed on whether that method chain could be run and cited, not on shipping the vehicle's full business scope. We do not report cycle-time or defect metrics for the vehicle here; those belong to the companion measurement manuscript.

Evidence layer: bounded evidence. The evidence layer contained maturity and effectiveness materials and stated their signal types. Workshop judgments were expert estimates and the materials were not presented as platform-measured return. For this path paper, the Evidence layer establishes that evidence boundaries were available to the closeout review; it is not used to introduce the companion paper's quantitative findings.

E. Phase-One Explore Closeout Decision

At the Explore closeout executive readout, management applied five questions: whether the playbook was citable; whether opportunity assets were countable; whether executable readiness was labeled ready or stub; whether effectiveness and maturity signals carried boundaries; and whether the next-stage path could be discussed concretely. The result was a pass for the Explore closeout. The *method chain* had run end-to-end under local facilitation (not a fully runnable automation chain; the Executable main path remained largely stubbed), the initial playbook and countable opportunity assets existed, and bounded signals could be reported.

The same review did *not* pass Scale admission. Replicability and seed coaches had not yet been demonstrated, the principal Executable execution chain remained largely stubbed, and the role, starter-kit, sponsor-cadence, and resource prerequisites

[1] Stage 0–5 table: `method-extract.md` under the Data Availability URL.

TABLE II
ORG-A EXPLORE METHOD–EVIDENCE ASSET LEDGER AT CLOSEOUT

Class	Asset class	Observed outcome	Readiness boundary / Consolidation need
Method	Method playbook / operations handbook	Version 1.0 was frozen, citable, and sufficient for the Explore closeout (public method card released; full handbook withheld).	It was not playbook v2; clauses that read well but do not run remained targets for revision.
Opportunity	Opportunity specifications and best practices	The release set contained 11 best practices and 13 specifications across fourteen themes; themes #5 and #9 remained observation items.	Countability did not mean 1:1 theme–spec–practice pairing or full convergence of every theme.
Executable	Executable plugins, skills, and recipes	The formal pack, including the `/sdd` path, skills, recipes, and adapters, was catalogued but largely stubbed on the main chain.	**A stub is not production-ready.** Main-chain hardening was Stage-2 work.
Evidence	Maturity/effectiveness evidence pack	The handoff included evidence pointers and signal-type boundaries.	The signals were process observations and expert estimates, not platform-measured return; this paper does not reuse their numeric results.

were not all in place. This was not a contradictory verdict. Explore closeout answered whether the organization had enough evidence to harden a feasible method; Scale admission would answer whether that hardened method could be transferred to another team.

The two-part decision is the practical value of the path. Without distinct closeout and admission decisions, management would have had to label the pilot either a failure because it was not ready for broad diffusion, or a scale success because it completed its planned activities. The stage-path account preserves the observed Explore achievement and the unresolved admission work at the same time.

F. Consolidation Midpoint; Scale Entry Remained Conditional

After Explore closeout, Org-A entered Consolidation for an indicative one-to-two-month hardening window (stage duration, not a calendar gate). The active work was framed as upgrading playbook v1 toward a standardized v2 and hardening the Executable main chain. Seed coaches and "everyone can teach" were capability targets and the Stage-2 exit criterion; they were not reported as achieved at Explore closeout.

Internal coach capacity was treated as a first-class anti-rebound control (failure mode 4): a Foundation / Practitioner / Seed-Coach ladder was in design, with courseware and platform work still pending at the midpoint. That shows scale enablement was being built for autonomous iteration after external facilitators step back—not that Org-A had completed training waves or met Scale admission.

The resulting posture is bounded. Claims 1–2 held (stages defined; Explore closeout met). Claim 3 remained conditional: Scale admission still needed team-can-teach, named owners, a hardened starter kit, and sponsor support. No new-team self-run, wave rollout, or organization-wide adoption was evidenced at the freeze point.

A Consolidation midpoint observation showed the client team operating selected method-pack practices on real demand without on-site consultants. Over a five-week cumulative slice, public process counts recorded 15 requirements, 18 changes, and 8 finished changes (process signals, not ROI). All 18 changes produced specifications via the mandatory `/sdd` path. Honest gaps remained in the method-pack Align/Verify reviews (not organizational Explore-closeout / Scale-admission decisions): verification (Verify) had human sign-off but no Reject/Conditional records yet; spec-review (Align) duration ran above playbook expectation; one AI-orchestration role was still hardening. Self-correction had started (client-run retrospective, toolchain self-fix, earlier estimation, Playbook v2 revision list) but was not finished. Scale admission remained unmet; Consolidation exit was likewise not yet claimed.

Org-A thus illustrates why movement from pilot evidence toward conditional Scale is a managed transition: Explore feasibility and assets, a client-operated Consolidation midpoint with disclosed gaps, and explicit negative evidence that kept Scale conditional. Later Scale admission can reuse those same declared conditions without rewriting the Explore result.

V. CASE ORG-B

A. Industrial Context and Evidence Basis

Org-B is an anonymized manufacturing enterprise. Unlike Org-A's multi-team agile portrait, Org-B presents a *harder process fit*: incumbent Integrated Product Development (IPD) *and* SAFe-class scaled-agile constraints on cross-team coordination. The same SDD-centered path must therefore be adapted to IPD decision/lifecycle structure and remain usable where scaled agile frames multi-team work; Explore evidence here is strongest at the team layer. We do not claim an organization-wide IPD×SAFe×AI4SE rewrite, SAFe program certification, or a completed mapping of every IPD/SAFe ceremony onto Explore closeout / Scale admission.

The evidence basis is an author briefing aligned to the statement of work (SOW). There is no consultant-side workspace mirror: client work remains on the client intranet. Facts in this section are therefore SOW- and briefing-level, and we mark delivery status explicitly.

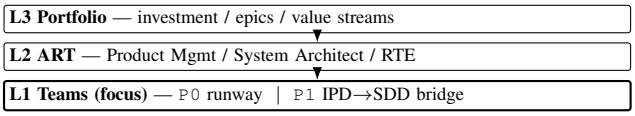


Fig. 6. Org-B SAFe AI-Native three-layer abstraction (Explore fit). L1 emphasis: P0 runway and the briefing-level P1 IPD bridge; L2/L3 remain compatibility surfaces.

B. Adoption After Infeasible Alternatives

During an initial co-located alternatives period, other consulting approaches led early exploration. Org-B concluded that those AI4SE transformation approaches were not feasible for its setting and selected the transformation path instantiated at Org-A, including its admission rhythm, as the primary Explore method. We report this as a client decision outcome. We do not name the alternative providers, and we do not present a competitive scorecard.

After that method switch, this team has been guiding multiple pilot teams under the path. At author briefing, Org-B was in late Explore (nearing closeout; not written as a formal closeout pass). Its stated intent is more hurried than Org-A: harden current pilots while sampling additional teams inside the Explore wave (multi-pilot Explore, not an admitted Scale wave). We record that as Explore sampling intent, not Scale admission or org-wide rollout.

C. IPD Layering, SAFe-Class Constraints, and the P1 Bridge

For SAFe-class fit, Explore used an abstract three-layer AI-Native SAFe reading (Fig. 6): Portfolio; ART; Teams. Focus is L1: **P0** runway enablement and the **P1** IPD-to-SDD bridge under Align/Verify/Merge. L2/L3 stay compatibility surfaces—not SAFe certification or org-wide rewrite.

IPD at P1 (briefing-level focus): RR is the raw requirement entering the local process. IPD requirement work items then branch by product type: equipment-oriented work follows SF→IR→SR→AR, whereas standalone software follows SF→IR→US. Here IR, SR, AR, and US denote initial, system, assigned, and user-story work items, respectively. These are requirement types and decomposition links, not stage gates. The mapping is Org-B's local briefing adaptation, not a universal IPD standard. The resulting items are packaged as SDD change requests and governed by Align/Verify/Merge. Harness Skills are intended to help analysts without deep IPD fluency meet both IPD and SDD constraints; we report the packaging design, not a controlled fluency study. P0 names runway enablement beside this P1 bridge. P2–P5 are not analyzed as completed phases. This answers Q3 as a late-Explore L1 adaptation design, not closeout, Consolidation, Scale, or an enterprise IPD/SAFe rewrite.

D. Coach Ladder, Multi-Mode Pilots, and Claim Bound

Anti-rebound coach ladder (external→gold seeds→seed coaches) is under construction; maturity/metric pack is Scale-prep only (no Org-B scores/DORA here). Explore samples heterogeneous modes (embedded; internal apps; large enterprise; AI-agent) while IPD/SAFe fit runs—mode diversity within Explore, not rollout. Harness stays open multi-SDD (OpenSpec / Superpowers / Spec Kit *class*) with MCP/CLI; method/harness fit work remains open at freeze, while coach training and measurement design are Scale-prep items, not observed Scale outcomes. Org-B therefore supports a harder-fit late-Explore adaptation account after a method-led switch; not Org-A-style Consolidation or Scale admission.

TABLE III
ORG-A AND ORG-B ON THE SAME TRANSFORMATION PATH AND RHYTHM

	Org-A	**Org-B**
Stage posture	∼6-week Explore finished; Consolidation midpoint (client-operated); Scale admission held	Late Explore / nearing closeout; multi-pilot Explore sampling (Scale admission not claimed)
Process fit	Multi-team agile; retained quality controls	IPD + SAFe-class scaled agile (harder compatibility)
Pilot shape	Enablement plus validation vehicle	Multi-mode; SAFe L1; P0 runway + P1 RR→IPD→SDD
Adoption	Sponsored enablement; Explore closeout met / Scale admission held	Method switch after client-judged infeasible alternatives
Harness	Playbook and plugin pack in handoff; main chain stubbed	Extensible multi-SDD harness plus MCP/CLI; not locked to one stack
Evidence depth	In-repo ledger and stage packet	Author briefing; no intranet mirror

VI. DISCUSSION

A. Reading the Two Cases Together

Org-A and Org-B occupy different positions on the same framework path (Table III) with unequal evidence depth: Org-A is the primary case (Explore done; Consolidation midpoint; Scale held; in-repo ledger), whereas Org-B is a briefing-level, harder-fit late-Explore adaptation (IPD + SAFe L1; P1 bridge), not a second completed Consolidation packet or Scale admission.

Org-A passed Explore closeout because the method chain had been exercised in real delivery, a citable playbook and opportunity inventory were available, and the ledger made remaining executable gaps explicit. The same evidence could not support Scale admission because the main executable chain, teachable ownership, starter kit, and seed-coach conditions were not yet in place. The two decisions therefore answer different uncertainty questions rather than contradicting one another.

The Consolidation midpoint adds process evidence without changing that boundary. The five-week slice shows repeated use of selected practices and mandatory `/sdd` entry, while the missing Reject/Conditional Verify records, slower Align review, and unfinished orchestration role identify work that still belongs in Consolidation. For Org-B, the IPD-to-SDD bridge is the corresponding adaptation layer: it changes how the method enters the local process, but the briefing record does not establish that the full path has cleared the same closeout or admission tests.

Taken together, Org-A supplies observed method-chain and Consolidation evidence; Org-B documents a harder-fit P1 packaging design without establishing Org-A's ledger or formal closeout. The increment over Stage-Gate business-case gates [3] is the AI4SE admission contract: stubs stay Consolidation work, Scale can be held after Explore, and harder-fit evidence is not auto-promoted.

B. Manager Takeaways

Managers should freeze the process-shaped harness and paradigm p on one stream, require a citable method and readiness ledger before tool-only rollout, keep unmet Scale items visible, and before Scale require named owners, seed coaches, a runnable starter kit, and no apply-path self-certification. These are design implications, not case-validated prerequisites.

VII. Threats to Validity

External: Org-A is primary; Org-B is late Explore briefing, not a second Consolidation site. **Construct:** "Scale-ready" is criteria-and-asset readiness, not platform-measured return. **Observer bias:** authors facilitated both engagements; mitigations are frozen evidence pointers, published negative evidence, and client Advance/Hold dispositions (Org-B: SOW-aligned briefing, no intranet mirror).

VIII. Related Work

We follow case-study guidance [4] and treat the path as a designed enablement artifact [5] at Org-A, not saturation or a completed DSR evaluate cycle. SPACE [1] supports refusing pilot slogans as Scale authorization; maturity measurement is the companion unit. Cynefin [6], SE layering [9], harness writing [11], and Thoughtworks retreat taxonomies [7], [8] are staging vocabulary; SDD/agentic writing [10] and AI-assisted maturity models [13] are peer/maturity context, while our unit is a path with closeout/admission against Stage-Gate business-case gates [3]. Those sources motivate staged governance but do not specify the brownfield transfer contract that separates pilot closeout from wave-specific Scale admission—the gap this path-and-admission framing addresses, with strongest evidence at Org-A and Org-B as briefing-level adaptation.

IX. Conclusion

Q1. We answer Q1 by specifying an evidence-based AI4SE transformation framework and method: the path organizes Explore / Consolidation / conditional Scale; the admission contract distinguishes citable, runnable, and transferable readiness (stub $\neq$ ready); and calendar time may prompt review but cannot substitute for closeout or admission (Section III).

Q2. At Org-A, the completed Explore produced a Method/Opportunity/Executable/Evidence handoff with readiness labels. That evidence supported Explore closeout and entry into Consolidation: an observed midpoint recorded 15 requirements, 18 changes, and 8 finished changes, with all 18 changes entering through `/sdd`. It did not support Scale admission because the Executable main chain remained largely stubbed, the midpoint showed no Reject/Conditional Verify records and longer-than-expected Align time, and the AI-orchestration role was still hardening (Section IV).

Q3. Org-B's briefing documents a team-level adaptation design in late Explore under IPD and SAFe-class constraints: P0 runway enablement and an RR→IPD layers→SDD P1 bridge were in focus, while formal closeout, later-stage delivery, Consolidation, Scale admission, and ART/Portfolio rewrite remain unclaimed (Section V-C).

Together the cases support a bounded feasibility/transfer claim, not two equivalent validation sites: pilot completion may trigger evidence review, but without transfer evidence it is not Scale authorization. The transferable artifact remains the path-plus-admission contract.

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Data Availability

Anonymized pack (CC-BY-4.0): https://github.com/ai4se-works/artifacts/tree/main/manuscripts/stage-gated-ai4se-path/data (stage criteria, Method–Evidence ledger, stub map, Org-A timeline/CSVs). Playbooks, plugins, and Org-B workspaces remain confidential; maturity tables are companion-only.

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